

# SPENDWISE - DEFINE PROBLEM STATEMENTS

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**Version:** 1.0.0 | **Date:** March 7, 2026 | **Status:**  Validated

## 1. USER PAIN POINTS

### **Pain Point 1: No Centralized Expense Tracking**

**Problem:** Users track expenses across multiple platforms

- Spreadsheets (manual entry)
- Paper receipts (easily lost)
- Mental notes (forgotten expenses)
- Bank statements (hard to categorize)

**Solution Implemented:** Single platform with all transactions in one place

**API Endpoint:** POST /api/transactions - Stores all transactions with user ID

### **Pain Point 2: No Spending Visibility**

**Problem:** Users don't know where money goes

- No category-wise breakdown
- No monthly summaries
- No spending patterns identified

**Solution Implemented:** Real-time analytics and category-based reports

**API Endpoint:** GET /api/transactions/stats/summary - Returns category breakdown

### **Pain Point 3: Manual Calculations**

**Problem:** Users manually calculate balances

- Error-prone calculations
- Time-consuming process

- No real-time updates

**Solution Implemented:** Automatic balance calculation with each transaction

**Implementation:** Server-side aggregation of income vs expense

### **Pain Point 4: No Budget Awareness**

**Problem:** Users overspend unknowingly

- No spending alerts
- No budget limits
- No savings goals tracking

**Solution Implemented:** Category-based tracking for future budget features

## 2. USER PERSONAS

### **Persona 1: Student Sarah**

|                             |                                                   |
|-----------------------------|---------------------------------------------------|
| <b>Age:</b>                 | 22                                                |
| <b>Occupation:</b>          | University Student                                |
| <b>Pain Points:</b>         | Limited income, needs to track expenses carefully |
| <b>Goals:</b>               | Save money, avoid overspending                    |
| <b>How Spendwise Helps:</b> | Track every expense, see where money goes         |

### **Persona 2: Professional Priya**

|                     |                                         |
|---------------------|-----------------------------------------|
| <b>Age:</b>         | 35                                      |
| <b>Occupation:</b>  | Software Engineer                       |
| <b>Pain Points:</b> | Busy schedule, needs automated tracking |
| <b>Goals:</b>       | Investment planning, tax preparation    |

**How Spendwise Helps:**

Quick entry, categorized transactions for tax time

 **Persona 3: Retiree Raj**

|                             |                                                        |
|-----------------------------|--------------------------------------------------------|
| <b>Age:</b>                 | 65                                                     |
| <b>Occupation:</b>          | Retired                                                |
| <b>Pain Points:</b>         | Fixed income, needs budget adherence                   |
| <b>Goals:</b>               | Track medical expenses, manage pension                 |
| <b>How Spendwise Helps:</b> | See if spending exceeds income, track medical category |

### 3. USER STORIES (IMPLEMENTED)

**Feature: User Registration** Scenario: New user signs up Given user provides name, email, password When password is hashed with bcrypt Then JWT token is generated And user data is stored in MongoDB

**Feature: User Login** Scenario: Existing user logs in Given user has registered before When correct email and password are provided Then JWT token is returned And user can access protected routes

**Feature: Add Transaction** Scenario: User adds expense Given user is authenticated with JWT When transaction details are provided Then transaction is saved with user ID And statistics are updated

**Feature: View Reports** Scenario: User views spending summary Given user has transactions When requesting statistics endpoint Then income, expense and balance are returned And category breakdown is provided

### 4. SUCCESS METRICS

| Metric            | Current Status | Measurement Method      |
|-------------------|----------------|-------------------------|
| API Response Time | ✅ <100ms       | Server logs             |
| User Registration | ✅ Working      | POST /api/auth/register |
| User Login        | ✅ Working      | POST /api/auth/login    |

|                      |           |                        |
|----------------------|-----------|------------------------|
| Transaction Creation | ✓ Working | POST /api/transactions |
| Error Rate           | ✓ <1%     | Exception logging      |
| Database Connection  | ✓ Stable  | Mongoose connection    |

## 5. PROBLEMS RESOLVED DURING DEVELOPMENT

| Problem                     | Solution                            | Status     |
|-----------------------------|-------------------------------------|------------|
| Port 5000 already in use    | Changed to port 5001 in .env        | ✓ Resolved |
| Database name case conflict | Changed to lowercase 'spendwise'    | ✓ Resolved |
| MODULE_NOT_FOUND errors     | Installed all dependencies          | ✓ Resolved |
| Cannot connect to server    | Used 127.0.0.1 instead of localhost | ✓ Resolved |

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Spendwise Problem Definition Document | Version 1.0.0 | March 7, 2026